—recognize the inevitability of the conclusions we have drawn. Quantum mechanics is different. By 1928 or so, many of the mathematical formulas and rules of quantum mechanics had been put in place and, ever since, it has been used to make the most precise and successful numerical predictions in the history of science. But in a real sense those who use quantum mechanics find themselves following rules and formulas laid down by the "founding fathers" of the theory—calculational procedures that are straightforward to carry out—without really understanding why the procedures work or what they really mean. Unlike relativity, few if any people ever grasp quantum mechanics at a "soulful" level.

What are we to make of this? Does it mean that on a microscopic level the universe operates in ways so obscure and unfamiliar that the human mind, evolved over eons to cope with phenomena on familiar everyday scales, is unable to fully grasp "what really goes on"? Or, might it be that through historical accident physicists have constructed an extremely awkward formulation of quantum mechanics that, although quantitatively successful, obfuscates the true nature of reality? No one knows. Maybe some time in the future some clever person will see clear to a new formulation that will fully reveal the "whys" and the "whats" of quantum mechanics. And then again, maybe not. The only thing we know with certainty is that quantum mechanics absolutely and unequivocally shows us that a number of basic concepts essential to our understanding of the familiar everyday world fail to have any meaning when our focus narrows to the microscopic realm. As a result, we must significantly modify both our language and our reasoning when attempting to understand and explain the universe on atomic and subatomic scales.

In the following sections we will develop the basics of this language and describe a number of the remarkable surprises it entails. If along the way quantum mechanics seems to you to be altogether bizarre or even ludicrous, you should bear in mind two things. First, beyond the fact that it is a mathematically coherent theory, the only reason we believe in quantum mechanics is because it yields predictions that have been verified to astounding accuracy. If someone can tell you volumes of intimate details of your childhood in excruciating detail, it's hard not to believe their claim of being your